



Department of CSE- Data Science

Major Project Presentation

Project Name

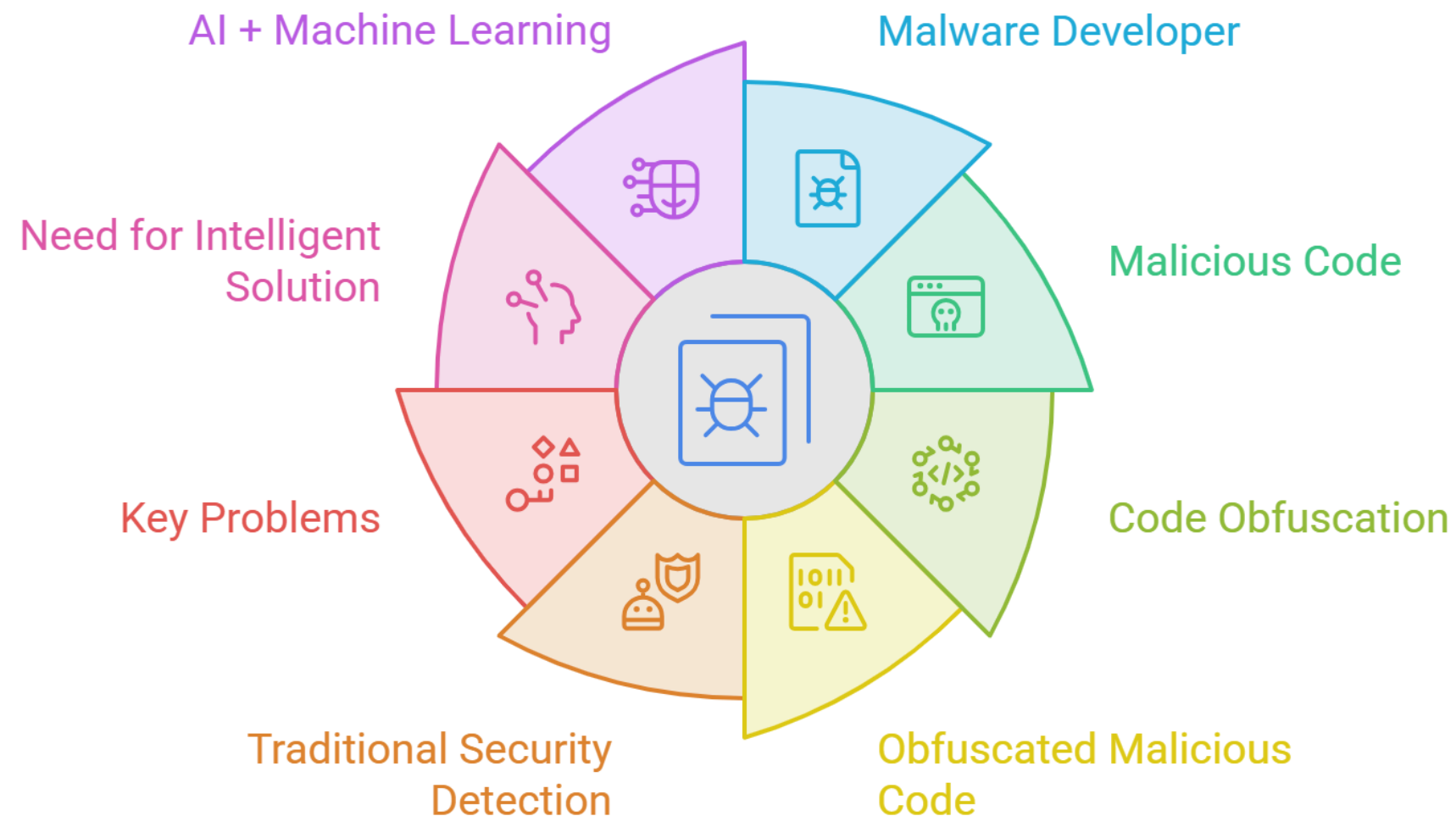
AI-Based Detection and Analysis of Obfuscated Malicious Code

“Leveraging Artificial Intelligence to uncover hidden malicious patterns in obfuscated code for smarter and more effective cybersecurity.”

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PROBLEM STATEMENT

AI-Driven Obfuscated Code Detection



- Malware developers use code obfuscation to hide malicious behavior and evade traditional security tools.
- Obfuscated or encoded scripts are difficult to analyze using conventional signature-based detection.
- Existing approaches may fail to detect new or previously unseen malicious patterns.
- Manual analysis of suspicious code is time-consuming and requires cybersecurity expertise.
- Therefore, an intelligent system is required to analyze obfuscated code and automatically identify malicious patterns.

Core Problem :

How can AI and Machine Learning be used to detect malicious behavior hidden inside obfuscated code?

PROPOSED SOLUTION

MACHINE LEARNING & DEEP LEARNING

Supervised ML classifiers (Random Forest, SVM, XGBoost) and Deep Learning models (CNN, LSTM) trained on labeled code samples to detect malicious patterns with high accuracy and generalization.

STATIC CODE ANALYSIS & FEATURE ENGINEERING

Code is analyzed statically without execution, extracting structural and behavioral features such as token frequency, code entropy, control flow patterns, and API call sequences.

CNN & LSTM MODELS

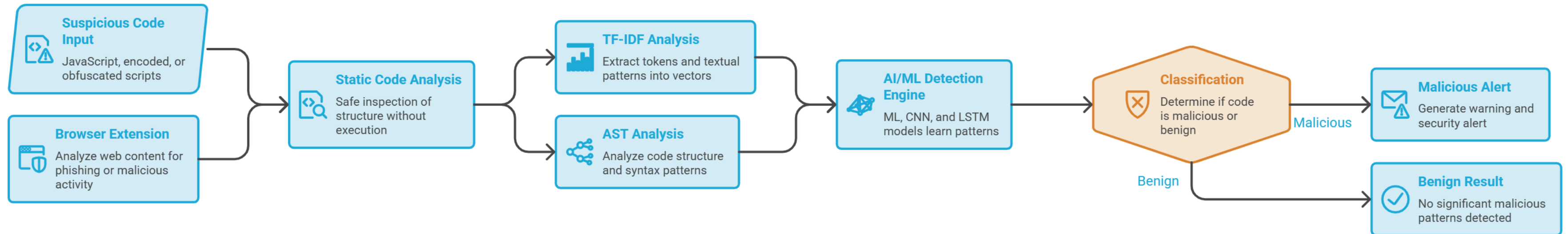
CNN captures local code patterns; LSTM models sequential code behavior. SMOTE (Synthetic Minority Oversampling) addresses class imbalance for robust, unbiased malware classification.

TF-IDF VECTORIZATION & AST ANALYSIS

TF-IDF converts code tokens into numerical feature vectors. Abstract Syntax Tree (AST) analysis captures structural code relationships to identify obfuscated and suspicious script patterns.

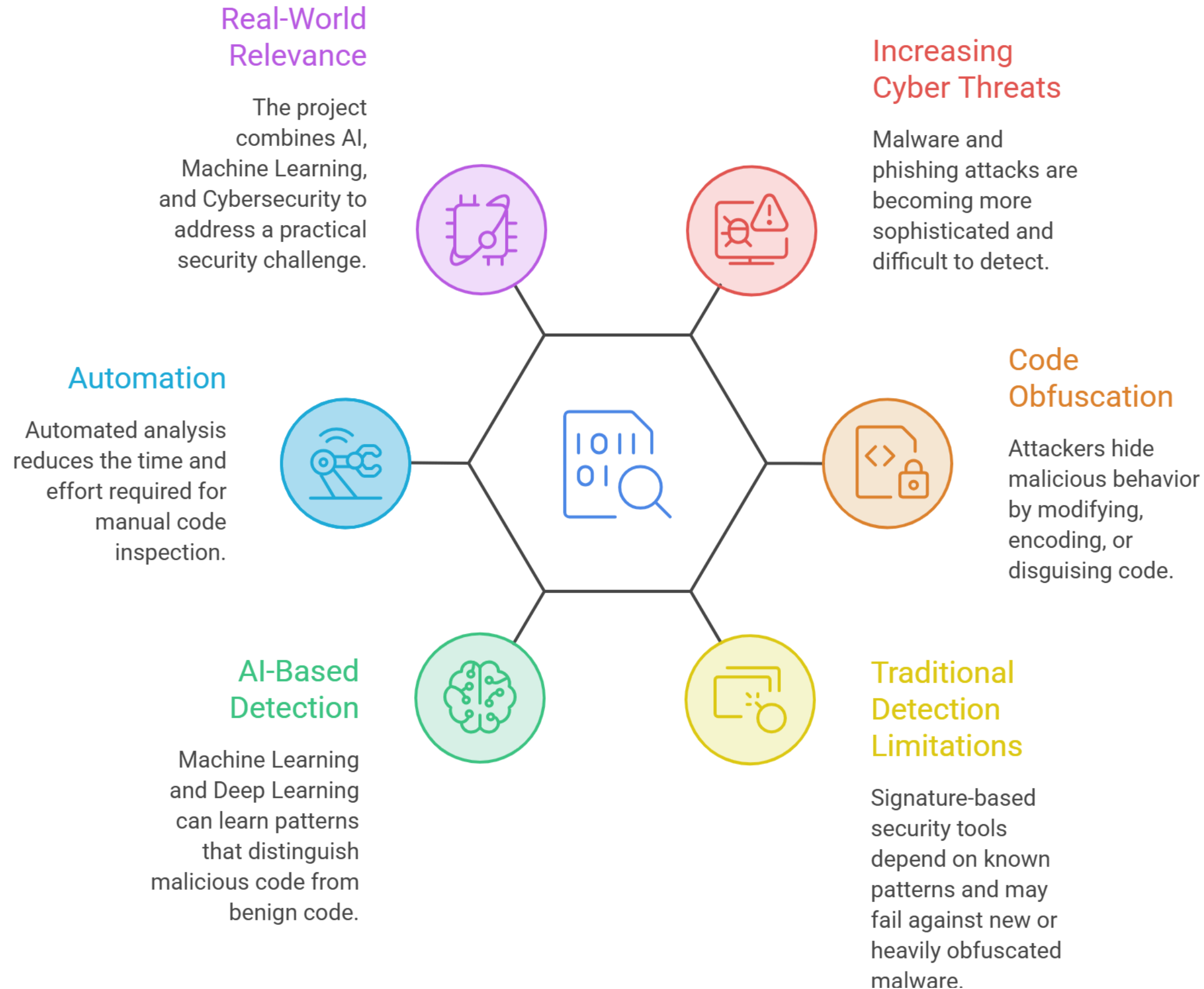
PROPOSED SOLUTION

AI-Based Malicious Code Deobfuscation and Detection Workflow

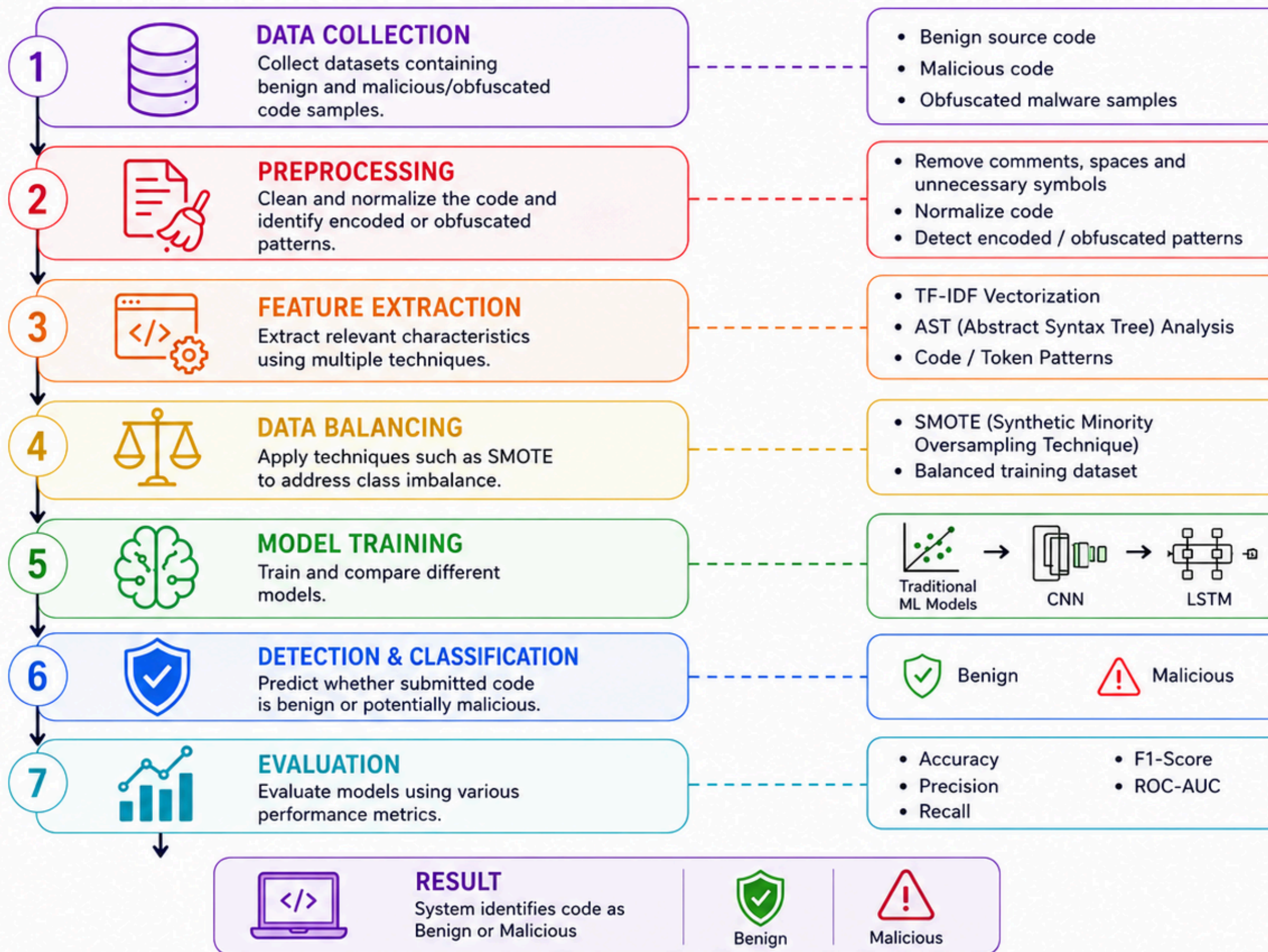


“An AI-powered system that uses static analysis, feature extraction, and ML/DL models to detect and classify obfuscated malicious code without executing it.”

WHY THIS TOPIC?



PROPOSED METHODOLOGY



EXPECTED OUTCOMES



High Accuracy

Precise detection via ML/DL classifiers



Zero-Day Coverage

Generalizes to unseen obfuscation patterns



Faster Analysis

Automated pipeline cuts manual triage time



Balanced Results

SMOTE keeps classification fair and unbiased



Research Validated

Supports both associated research papers

- Accurate detection of obfuscated malicious code with high precision and recall using ML/DL classifiers.
- Ability to identify previously unseen, zero-day obfuscation patterns through generalized feature learning.
- Faster, automated static analysis pipeline that reduces manual effort needed for malware triage.
- Robust, unbiased classification despite class imbalance, achieved through SMOTE-balanced training.
- Validated research contribution, supporting publication of the two associated research papers.

Key Impact :

A reliable, scalable AI system that detects hidden malicious code faster and more accurately than traditional tools.

Thank You